Venkata Lakshmi Parimala Pasupuleti

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SUMMARY

Senior software engineer with **9+ years shipping production systems** and deep hands-on ownership of GenAI agents, evaluation, and enterprise compliance. At Intuit, **builds and operates production Generative AI workflows** end to end: tool calling, orchestration, state management, RAG, structured outputs, and LLMOps instrumentation. At Entrata (100K+ MAU regulated SaaS), owned **PHI/HIPAA data infrastructure**, AWS production operations (S3, EC2, RDS, Lambda, IAM), Terraform infrastructure as code, and GitHub Actions CI/CD - shipping and operating production services, not editing templates. Built **Leasel**: an 8-agent LangGraph system with human-in-the-loop controls, provenance on AI outputs, and agent observability (93.6% precision across 1,240 test cases). Strong in Python, React, SQL, REST APIs, and the full AWS/Terraform/GitHub Actions stack for regulated environments. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

GenAI Agents & Workflow Orchestration | Tool Use / State Management / Tool Execution | Human-in-the-Loop & Agent Observability | LLM Evaluation & LLMOps | Python / React / SQL / REST APIs | AWS Production Operations (IAM, S3, EC2, RDS, Lambda) | Terraform / GitHub Actions CI/CD | PHI / HIPAA / Enterprise Compliance | Authentication & Authorization | Observability / Logging / Monitoring | Enterprise System Integrations & APIs | Root Cause Analysis & Production Debugging

EXPERIENCE

Intuit

Jun 2026 -- Present

Software Engineer - GenAI & AI Systems - Virginia, USA

- **Builds and operates production Generative AI agents and workflows**: tool calling, orchestration with LangGraph, state management, RAG, context management, structured outputs, and interaction with enterprise financial data systems.
- Owns **deployment and operation on AWS** with Terraform infrastructure as code and GitHub Actions CI/CD - including environment management, secrets, IAM, and repeatable promotion to production.
- Implements **evaluation, observability, and human-in-the-loop controls**: LLM-as-judge harnesses, golden datasets, provenance on AI outputs, and instrumentation to validate before moving up the autonomy ladder.
- Implements **authentication, authorization, logging, monitoring, and error handling** for sensitive financial data; adds guardrails and fallbacks around every LLM call; meets enterprise security and auditability requirements.
- Sits with business stakeholders to scope work: translates ambiguous operational problems into concrete services, APIs, and UI changes.

Entrata - Multifamily & Affordable Housing SaaS

Oct 2020 -- Jul 2024

Senior Software Engineer - 100K+ monthly active users · Python, React, AWS, Terraform

- Owned **PHI document infrastructure end to end**: AWS S3 migration with Libsodium encryption, HIPAA-compliant access controls, audit logging, and authorization enforcement - meeting enterprise security and compliance requirements in a regulated environment.
- Built and operated **production workloads on AWS**: EC2, RDS, S3, Lambda, SQS, IAM - owning networking, identity, operational concerns, and incident response, not editing templates.
- Owned **Terraform infrastructure as code and GitHub Actions CI/CD**: Docker microservices architecture (50% better scalability, 30% faster deployments), with full environment promotion and secret management.
- Integrated **enterprise and legacy systems via REST APIs**: external federal data sources (HUD, Tax Credit), event streams (RabbitMQ async pipeline), and third-party data platforms using versioned contracts and non-invasive integration patterns.
- Managed 20+ concurrent feature workstreams across 120+ Agile sprints; reviewed 300+ PRs; mentored junior engineers; built Python/Bash automation eliminating 80% of manual operational work.

Entrata India

Sep 2018 -- Oct 2020

Junior Software Engineer - Pune, India

- Built production Python/PHP/PostgreSQL modules for regulated housing workflows; led AWS S3 cloud migration from legacy storage.

PROJECTS

Leasel - Production-Grade 8-Agent AI System

Python · LangGraph · FastAPI · RAG · Human-in-the-Loop · Agent Observability

End-to-end AI agent system for housing lease compliance: **tool execution, state management, orchestration** (LangGraph), RAG retrieval, and **human-in-the-loop escalation** for consequential decisions - the LLM never makes final eligibility or financial calls. Full evaluation harness (LLM-as-judge + golden datasets): 93.6% precision, 87.9% F1 across 1,240 test cases. Provenance on all material outputs; confidence scoring gates autonomy level. Demonstrates production-grade GenAI agent engineering with enterprise-level controls.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - LLMOps Evaluation & Observability Toolkit

Python · FastAPI · 5 LLM Providers · LLMOps · CI/CD

15-endpoint FastAPI service with **agent observability and LLMOps** instrumentation across five LLM providers. Evaluates accuracy, hallucination rate, retrieval quality, and compliance with CI regression tests and golden datasets. Built as a reusable evaluation substrate - the same pattern Sedgwick needs across multiple AI build streams.

github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU Anantapur — 2017